

MUHAMMAD KASHIF LATIF

SOFTWARE ENGINEER | FULL-STACK DEVELOPER | AI/ML ENGINEER

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PROFESSIONAL PROFILE

Computer Science student with a 3.91/4.00 GPA and hands-on experience across AI engineering, full-stack development, software engineering, backend systems, and application development through three remote technical internships. Built and deployed role-based management platforms, e-commerce products, authentication-driven applications, AI recommendation systems, chatbots, real-time communication software, and machine-learning models for classification, regression, clustering, and prediction. Comfortable moving from problem definition and data/model logic to responsive frontend interfaces, APIs, databases, testing, Git-based workflows, and cloud deployment using Python, JavaScript/TypeScript, Next.js, React, Node.js, Express.js, PostgreSQL, Supabase, MongoDB, and Scikit-learn.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C, C++, HTML5, CSS3

Frontend: React.js, Next.js, Tailwind CSS, Bootstrap, responsive UI, component-based development, state management

Backend & Databases: Node.js, Express.js, REST APIs, CRUD, authentication, role-based access, PostgreSQL, MySQL, MongoDB, Mongoose, Supabase, Oracle

AI/ML: Classification, regression, clustering, recommendation systems, NLP-style text classification, feature engineering, model training and evaluation

Models & Libraries: Naive Bayes, SVM, Logistic Regression, Decision Trees, Random Forest, Linear Regression, K-Means, TF-IDF, cosine similarity, Scikit-learn, Pandas, NumPy, Matplotlib, PyTorch

Tools & Deployment: Git, GitHub, Vercel, API integration, Generative AI, LLM/chatbot integration, testing and deployment workflows

EXPERIENCE

AI Engineer Intern | DecodeLabs | Remote | 2-month internship

- Built a Python rule-based AI chatbot and developed a practical ML portfolio covering email/text classification, student pass/fail prediction, customer purchase prediction, heart-disease classification, employee-attrition prediction, electricity-bill estimation, and stock-market clustering.
- Worked through data preparation, feature representation, model training, prediction logic, and experimentation with Naive Bayes, SVM, Logistic Regression, Random Forest, Decision Trees, Linear Regression, and K-Means.

Full-Stack Developer Intern | CodeAlpha | Remote | 2-month internship

- Built and deployed production-style web applications including Lumina Luxe, a large jewelry e-commerce platform; NexWalk, an additional e-commerce experience; and DishCraft, an AI-powered food recommendation application.
- Implemented responsive interfaces, application flows, data-backed features, authentication-oriented experiences, product discovery, recommendation logic, testing, Git/GitHub workflows, and Vercel deployment using modern JavaScript/TypeScript technologies.

App Developer Intern | HexSoftware | Remote | 2-month internship

- Developed customer-facing applications and login/authentication flows with structured navigation, form handling, validation, reusable interface components, user-focused design, testing, and deployment.
- Strengthened end-to-end application-development skills by translating functional requirements into usable software experiences rather than isolated demos.

SELECTED ENGINEERING & AI PROJECTS

COMCAT University Management Portal | [Next.js](#), [TypeScript](#), [PostgreSQL](#), [Supabase](#), [NextAuth](#) | [Live](#) / [GitHub](#) — Full-stack academic administration system with separate Admin, Teacher, and Student portals; role-based authentication; student/teacher/subject management; attendance, fees, admissions, enrollments, announcements, CRUD/API workflows, and an integrated AI chatbot.

Lumina Luxe E-Commerce Platform | [Next.js](#), [TypeScript](#), [Supabase](#), [PostgreSQL](#) | [Live](#) / [GitHub](#) — Deployed jewelry e-commerce platform with a 700+ item catalog, search and product discovery, persistent cart/wishlist, authentication, checkout and order flows, customer interactions, analytics, and an admin dashboard for catalog and order operations.

DishCraft AI Recommendation Engine | [JavaScript](#), [TF-IDF](#), [Cosine Similarity](#) | [Live](#) / [GitHub](#) — Content-based recommendation system that converts food preferences into weighted features, computes similarity scores, ranks Top-N recipe matches, and handles cold-start or sparse-input cases through fallback logic.

Classification & Predictive ML Portfolio | [Python](#), [Scikit-learn](#), [Pandas](#), [NumPy](#) | [GitHub](#) — Built supervised-learning projects for Naive Bayes email/text classification, SVM student pass/fail prediction, Logistic Regression customer-purchase prediction, Random Forest heart-disease classification, and employee-attrition prediction using structured training and prediction workflows.

Regression & Forecasting Models | [Python](#), [Linear Regression](#), [Decision Trees](#) | [GitHub](#) — Developed electricity-bill prediction projects that transform household or appliance-use inputs into estimated monthly cost outputs, comparing regression approaches and visualizing model behavior with Matplotlib.

K-Means Market Pattern Clustering | [Python](#), [K-Means](#), [Unsupervised Learning](#) | [GitHub](#) — Applied K-Means clustering to daily stock-market price and volume data to discover hidden behavioral groupings and practice unsupervised pattern discovery beyond labeled classification tasks.

Rule-Based AI Chatbot | [Python](#), [Conversational Logic](#) | [GitHub](#) — Created a command-line intelligent chatbot using rule-based intent/response logic, structured conversation flows, Python control structures, and deterministic response handling as a foundational AI system.

Advanced Real-Time Communication System | [Python](#), [Sockets](#), [Multithreading](#) | [GitHub](#) — Engineered a multi-user client-server application supporting public/private messaging, rooms, file sharing, and audio/video communication using socket programming and concurrent server handling.

Trust Mark Real Estate Advisor | [Next.js](#), [TypeScript](#), [Tailwind CSS](#) | [Live](#) / [GitHub](#) — Delivered a polished client-facing real-estate website for presenting listings, investment opportunities, company information, and responsive customer journeys with a professional interface and deployment-ready frontend structure.

NexWalk & DigiWallet Web Applications | [TypeScript](#), [Next.js](#), [JavaScript](#) | [Live](#) / [GitHub](#) — Built additional product-oriented applications covering e-commerce-style user experiences and a digital-wallet interface, strengthening reusable component design, client-side state, user flows, responsive layouts, and deployment practice.

GPT-Style Conversational AI Interface | [TypeScript](#), [Next.js](#), [AI API Integration](#) | [Live](#) / [GitHub](#) — Built and deployed a polished conversational interface for human-like AI interaction, focusing on chat UX, responsive presentation, message flows, and integration-ready frontend structure.

PDF AI Assistant | [Python](#), [Generative AI](#), [Gemini API](#) | [GitHub](#) — Developed an AI-powered application that accepts PDF documents and enables users to ask questions about uploaded content through a conversational interface backed by external generative-AI services.

Additional Engineering Portfolio | [Python](#), [JavaScript](#), [HTML/CSS](#) | [GitHub](#) — Includes a rule-based AI chatbot, multi-user authenticated chatroom, to-do application with local storage, JavaScript learning projects, portfolio websites, frontend experiments, and other public software/AI repositories demonstrating continuous hands-on development.

EDUCATION

BS Computer Science | Minhaj University Lahore | **GPA: 3.91/4.00** | Sep 2024 - Expected 2028

Intermediate (ICS) | Defence Degree College for Boys, DHA Phase 2 | A Grade | May 2021 - Jul 2023

Matriculation (Computer Science) | DHA Senior School for Boys, DHA Phase 2 | A* Grade | Mar 2019 - Apr 2021